

System

$$H_S = E_{MF} + \sum_k \psi_k^\dagger \begin{pmatrix} \eta_k + \xi_k & -\Delta \\ -\Delta & \eta_k - \xi_k \end{pmatrix} \psi_k$$

$$\psi_k = \begin{pmatrix} c_{a,k} \\ c_{b,k} \end{pmatrix} \quad (\text{proj } a_k, b_k)$$

$$\left. \begin{aligned} \eta_k &= \frac{\bar{\epsilon}_a + \bar{\epsilon}_b}{2} \\ \xi_k &= \frac{\bar{\epsilon}_a - \bar{\epsilon}_b}{2} \end{aligned} \right\} \begin{aligned} \eta_k + \xi_k &= \bar{\epsilon}_a = \epsilon_a(k) + V n_b \\ \eta_k - \xi_k &= \bar{\epsilon}_b = \epsilon_b(k) + V n_a \end{aligned}$$

$$E_{\alpha, k} = \eta_k \pm \sqrt{\xi_k^2 + \Delta^2}$$

$$\Delta = \sum_k \frac{V \cdot \Delta}{2 \sqrt{\xi_k^2 + \Delta^2}} (n_{p,k} - n_{d,k})$$

$$\Delta = \sum_k m_k \langle c_{a,k}^\dagger c_{b,k} \rangle$$

$$r_{\lambda, k} = \begin{pmatrix} d_k \\ p_k \end{pmatrix} \Big|_{\lambda \neq p}$$

$$\psi_k = \begin{pmatrix} u_k & v_k \\ -v_k & u_k \end{pmatrix} \chi_k$$

$$\rightarrow \frac{u_k^2}{v_k^2} = \frac{1}{2} \left(1 \pm \frac{\xi_k}{\sqrt{\xi_k^2 + \Delta^2}} \right)$$

$$\rightarrow u_k v_k = \frac{1}{2} \sin(\theta_k) = \frac{\Delta}{\sqrt{\xi_k^2 + \Delta^2}}$$



$$H_S = E_{MF} + \sum_k (E_{a,k} n_{a,k} + E_{b,k} n_{b,k}) \quad n_{\lambda, k} = \langle \psi_k^\dagger \hat{n}_{\lambda, k} \psi_k \rangle$$

Bath:

$$H_B = \sum_{\lambda, \mathbf{r}} \omega_{\lambda, \mathbf{r}} a_{\lambda, \mathbf{r}}^\dagger a_{\lambda, \mathbf{r}} \quad , \quad \omega_{\lambda, \mathbf{r}} = \sqrt{\omega_{\lambda, 0}^2 + c^2 |\mathbf{r}|^2} \quad (\text{harmon})$$

Interakcija:

od zadatka: N realni reprezentiraj: $\lambda = \{a, b\}$

$$H = \sum_{i,j,\lambda} \left(\epsilon^{(\lambda)} \delta_{ij} + t_{ij}^{(\lambda)} e^{-ig(a_i^\dagger + a_i)} \right) c_{\lambda,i}^\dagger c_{\lambda,j} + \text{h.c.} +$$

$$+ V n_b \sum_j c_{a,j}^\dagger c_{a,j} + V n_a \sum_j c_{b,j}^\dagger c_{b,j} - V \phi \sum_j c_{a,j}^\dagger c_{a,j} - V \phi^\dagger \sum_j c_{b,j}^\dagger c_{b,j} + E_{MF}$$

$$\rightarrow c_{\lambda,ij}^\dagger = \frac{1}{\sqrt{N}} \sum_k e^{ikr_{ij}} c_{\lambda,k}^\dagger \rightarrow n_{\lambda} = \frac{1}{N} \sum_{k,p} e^{i(k-p)r_{ij}} \langle c_{\lambda,k}^\dagger c_{\lambda,p} \rangle = \frac{1}{N} \sum_k \langle c_{\lambda,k}^\dagger c_{\lambda,k} \rangle$$

$$\Delta = + \frac{V}{N} \sum_k \langle c_{a,k}^\dagger c_{b,k} \rangle$$

$$\rightarrow e^{ig(a_i^\dagger + a_i)} = e^{-ig a_i^\dagger} e^{-ig a_i} e^{\frac{[a_i^\dagger, a_i] g^2}{2}} \approx (1 - ig a_i^\dagger)(1 - ig a_i) \approx 1 - ig(a_i^\dagger + a_i)$$

$$\cdot \sum_{k,p} \frac{1}{N} e^{i(k-p)r_{ij}} \sum_{j,i,\lambda} \left(\epsilon^{(\lambda)} + t_{ij}^{(\lambda)} (e^{ik} + e^{-ik}) \right) c_{\lambda,k}^\dagger c_{\lambda,p} = \sum_{k,p} \left(\epsilon^{(\lambda)} + 2t^{(\lambda)} \cos(k) \right) c_{\lambda,k}^\dagger c_{\lambda,p}$$

s.v. interaksi :

$$a_j^\dagger = \frac{1}{\sqrt{N}} \sum_z e^{i2R_j} a_z^\dagger$$

$$-ig \sum_{j, \lambda} t^{(1)} \left(\frac{1}{\sqrt{N}} \sum_z \left(e^{i2R_j} a_z^\dagger + e^{i2R_j} a_z \right) \right) \left(\frac{1}{\sqrt{N}} \sum_{\lambda'} e^{i(k-p)R_j} e^{ik} c_{\lambda, k}^\dagger c_{\lambda', p} \right) + h.c. =$$

$$\sum_{\lambda} \frac{1}{N} \sum_z e^{i(k-p \pm z)} = \delta_{p, k \pm z}$$

$$= -i \frac{g}{\sqrt{N}} \sum_{\lambda, k, z} a_z^\dagger c_{\lambda, k} c_{\lambda, k+z} t^{(1)} \left(e^{ik} - \underbrace{e^{i(k+z)}}_{h.c.} \right) -$$

$$-i \frac{g}{\sqrt{N}} \sum_{\lambda, k, z} a_z c_{\lambda, k} c_{\lambda, k+z} t^{(1)} \left(e^{ik} - \underbrace{e^{i(k-z)}}_{h.c.} \right) =$$

$$= \sum_{\lambda, k, z} (a_z^\dagger + a_{-z}) \left[-i \frac{g}{\sqrt{N}} t^{(1)} (e^{ik} - e^{i(k+z)}) \right] c_{\lambda, k}^\dagger c_{\lambda, k+z}$$

$$G_\lambda(k, z)$$

band index

diagonal or λ : $c_a^\dagger c_a \alpha_i c_b^\dagger c_b$

$$\rightarrow H_{SB} = \sum_{\lambda, k, z} (a_z^\dagger + a_{-z}) G_\lambda(k, z) c_{\lambda, k}^\dagger c_{\lambda, k+z}$$

$$g_{\mu\nu}(k, z)$$

rotasi

$$c_{\lambda, k} = U_{\mu, \lambda} \gamma_{\mu, k} \quad \begin{pmatrix} c_{a, k} \\ c_{b, k} \end{pmatrix} = \begin{pmatrix} u & v \\ -v & u \end{pmatrix} \begin{pmatrix} \alpha_k \\ \beta_k \end{pmatrix}$$

$$\rightarrow G_\lambda(k, z) U_{\mu, \lambda}(k) U_{\nu, \lambda}(k+z) \gamma_{\mu, k}^\dagger \gamma_{\nu, k+z}$$

$$\rightarrow \lambda = a : c_{a, k}^\dagger c_{a, k+z} = \begin{pmatrix} u & v \\ -v & u \end{pmatrix} \begin{pmatrix} \alpha_k^\dagger + v \beta_k^\dagger \\ -v \alpha_k^\dagger + u \beta_k^\dagger \end{pmatrix} \begin{pmatrix} u & v \\ -v & u \end{pmatrix} \begin{pmatrix} \alpha_{k+z} \\ \beta_{k+z} \end{pmatrix} = \mu \mu' (\alpha_k^\dagger \alpha_{k+z}) + \mu v' (\alpha_k^\dagger \beta_{k+z}) + v \mu' (\beta_k^\dagger \alpha_{k+z}) - \mu v' (\beta_k^\dagger \beta_{k+z})$$

$$\rightarrow g_{aa}(k, z) = \mu_k \mu_{k+z} G_a(k, z) + v_k v_{k+z} G_b(k, z) \quad \alpha(k) \rightarrow \alpha(k+z)$$

$$\rightarrow g_{ab}(k, z) = \mu_k v_{k+z} G_a(k, z) - v_k \mu_{k+z} G_b(k, z) \quad \beta \rightarrow \alpha$$

$$\rightarrow g_{ba}(k, z) = v_k \mu_{k+z} G_a(k, z) - \mu_k v_{k+z} G_b(k, z) \quad \alpha \rightarrow \beta$$

$$\rightarrow g_{bb}(k, z) = v_k v_{k+z} G_a(k, z) + \mu_k \mu_{k+z} G_b(k, z) \quad \beta \rightarrow \beta$$

$$\rightarrow H_{SB} = \sum_{\lambda, k} (a_z^\dagger + a_{-z}) \left[g_{aa} \alpha_k^\dagger \alpha_{k+z} + g_{ab} \alpha_k^\dagger \beta_{k+z} + g_{ba} \beta_k^\dagger \alpha_{k+z} + g_{bb} \beta_k^\dagger \beta_{k+z} \right]$$

→ Lindblad:

$$L_{\mu\nu}^{(\mu\nu)} \propto \mu_k^\dagger \nu_{k+z} ; \{\mu, \nu\} = \{\alpha, \beta\}.$$

→ rate?

$$\gamma_i = 2\pi |\langle f | H_{SB} | i \rangle|^2 \delta(E_f - E_i)$$

$$H_{SB} = \sum_{k_2} \sum_{\mu\nu} \left(a_z^\dagger g_{\mu\nu}(k, z) \gamma_{\mu, k}^\dagger \gamma_{\nu, k+z} + a_z g_{\mu\nu}(k+z, -z) \gamma_{\mu, k+z}^\dagger \gamma_{\nu, k} \right)$$

① ABSORPTION formula: $|i\rangle = |N_z\rangle \otimes |1_{\nu, k} 0_{\mu, k+z}\rangle$

$$|f\rangle = |N_z-1\rangle \otimes |0_{\nu, k} 1_{\mu, k+z}\rangle$$

$$\underbrace{\langle N_z-1 | a_z | N_z \rangle}_{\sqrt{N_z}} \underbrace{\langle 0_{\nu, k} 1_{\mu, k+z} | \gamma_{\mu, k+z}^\dagger \gamma_{\nu, k} | 1_{\nu, k} 0_{\mu, k+z} \rangle}_{|0_{\nu, k} 1_{\mu, k+z}\rangle} g_{\mu\nu}(k+z, -z)$$

$$\mathcal{M}_1 = \sqrt{N_z} \cdot g_{\mu\nu}(k+z, -z)$$

② emission $k+z \rightarrow k$: $|i\rangle = |N_z\rangle |0_{\mu, k} 1_{\nu, k+z}\rangle$

$$|f\rangle = |N_z+1\rangle |1_{\mu, k} 0_{\nu, k+z}\rangle$$

$$\underbrace{\langle N+1 | a^\dagger | N \rangle}_{\sqrt{N_z+1}} \langle 1_{\mu, k} 0_{\nu, k+z} | \gamma_{\mu, k}^\dagger \gamma_{\nu, k+z} | 0_{\mu, k} 1_{\nu, k+z} \rangle g_{\mu\nu}(k, z) = \mathcal{M}_2^{\mu\nu}$$

?

Ans: $\gamma_{k_2}^{\alpha\beta} = 2\pi |g_{\alpha\beta}(k+z, -z)|^2 N_z (n_{\alpha, k} (1 - n_{\alpha, k+z})) \delta(E_{\alpha, k+z} - E_{\alpha, k} - \omega_z) +$
 $+ 2\pi |g_{\alpha\beta}(k, z)|^2 (N_z+1) (n_{\alpha, k+z} (1 - n_{\alpha, k})) \delta(E_{\alpha, k} - E_{\alpha, k+z} + \omega_z)$

abs, em

QF, rate?

Ans: $\gamma_{k_2}^{\alpha\beta} = 2\pi |g_{\alpha\beta}(k+z, -z)|^2 N_z (n_{\beta, k} (1 - n_{\beta, k+z})) \delta(E_{\alpha, k+z} - \dots) +$
 $+ 2\pi |g_{\alpha\beta}(k, z)|^2 (N_z+1) (n_{\alpha, k} (1 - n_{\alpha, k+z})) \delta(\dots) \quad ?$